

**KULSARY, Kazakhstan**WMO: **357150**Lat: **46.967N**Lon: **54.017E**Elev: **59**StdP: **14.66**Time Zone: **5.00 (E05)**Period: **82-94**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|------|------|--------------------------|------|-----|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%   |      |                          |      |     |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |     |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p) |
| 2             | -10.2      | -6.1 | -15.3                         | 2.4 | -8.8 | -11.7 | 2.9 | -5.0 | 45.7                  | 18.3 | 45.0 | 18.6 | 8.0                      | 50   | N/A |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 21.4                   | 100.3           | 67.0 | 96.6 | 66.3 | 93.2 | 65.2 | 71.1                | 88.4 | 69.4 | 87.3 | 68.1 | 85.3 | 13.3                 | 110  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 66.3                            | 97.0 | 75.9 | 64.0 | 89.6 | 76.8 | 61.8 | 82.6 | 75.9 | 34.9          | 87.8 | 33.6 | 86.4 | 32.4 | 84.9 | 87.3              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |      |      | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |     |    |    |
|-------------------|------|------|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|-----|----|----|
|                   |      |      | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |     |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p) |    |    |
| (4) 41.2          | 36.1 | 31.7 | N/A                        | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |    |    |
| (5)               |      |      | N/A                        | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |    |    |

**Monthly Climatic Design Conditions**

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 48.0   | 18.3  | 14.7  | 28.9  | 52.7  | 64.4  | 76.3  | 80.1  | 76.5  | 63.1  | 47.3  | 29.8  | 21.4  |  |  |
|                                                                        | DBStd       | 25.00  | 9.12  | 11.92 | 11.77 | 8.80  | 7.64  | 6.75  | 6.24  | 6.27  | 7.98  | 7.78  | 11.02 | 10.04 |  |  |
|                                                                        | HDD50       | 4340   | 984   | 989   | 656   | 71    | 2     | 0     | 0     | 0     | 5     | 141   | 607   | 885   |  |  |
|                                                                        | HDD65       | 7553   | 1449  | 1409  | 1120  | 381   | 106   | 1     | 1     | 4     | 128   | 547   | 1057  | 1350  |  |  |
|                                                                        | CDD50       | 3603   | 0     | 0     | 1     | 151   | 449   | 790   | 934   | 822   | 397   | 59    | 0     | 0     |  |  |
|                                                                        | CDD65       | 1342   | 0     | 0     | 0     | 11    | 88    | 342   | 470   | 361   | 70    | 0     | 0     | 0     |  |  |
|                                                                        | CDH74       | 16085  | 0     | 0     | 0     | 113   | 1108  | 4067  | 5970  | 4012  | 802   | 13    | 0     | 0     |  |  |
|                                                                        | CDH80       | 7828   | 0     | 0     | 0     | 15    | 393   | 2045  | 3209  | 1890  | 276   | 0     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 12.9   | 13.6  | 15.6  | 15.0  | 14.1  | 12.5  | 11.9  | 11.2  | 11.2  | 11.9  | 11.8  | 12.6  | 13.4  |  |  |
| (15) Precipitation                                                     | PrecAvg     | 6.1    | 0.4   | 0.4   | 0.5   | 0.7   | 0.6   | 0.9   | 0.5   | 0.2   | 0.2   | 0.5   | 0.6   | 0.6   |  |  |
|                                                                        | PrecMax     | 9.9    | 1.0   | 0.7   | 1.7   | 1.8   | 1.2   | 2.8   | 1.5   | 1.1   | 0.6   | 1.5   | 1.0   | 1.3   |  |  |
|                                                                        | PrecMin     | 3.3    | 0.0   | 0.1   | 0.0   | 0.0   | 0.1   | 0.0   | 0.0   | 0.0   | 0.0   | 0.1   | 0.1   | 0.2   |  |  |
|                                                                        | PrecStd     | 1.9    | 0.2   | 0.2   | 0.4   | 0.6   | 0.4   | 0.9   | 0.4   | 0.3   | 0.2   | 0.4   | 0.3   | 0.3   |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 37.1  | 39.6  | 59.2  | 81.9  | 92.1  | 103.4 | 105.3 | 99.9  | 91.4  | 76.1  | 52.5  | 41.7  |  |  |
|                                                                        |             | MCWB   | 34.7  | 37.2  | 47.6  | 57.5  | 61.2  | 67.6  | 68.3  | 67.9  | 61.3  | 55.3  | 47.1  | 39.2  |  |  |
|                                                                        | 2%          | DB     | 34.2  | 34.2  | 53.9  | 77.8  | 88.1  | 98.8  | 102.0 | 96.0  | 87.0  | 70.9  | 49.0  | 38.3  |  |  |
|                                                                        |             | MCWB   | 32.3  | 32.7  | 44.7  | 54.9  | 60.5  | 66.0  | 68.4  | 66.9  | 59.4  | 53.6  | 44.2  | 36.6  |  |  |
|                                                                        | 5%          | DB     | 33.0  | 32.7  | 49.4  | 73.0  | 84.4  | 94.9  | 98.0  | 92.8  | 83.0  | 65.4  | 45.9  | 35.6  |  |  |
|                                                                        |             | MCWB   | 31.5  | 31.2  | 42.2  | 53.7  | 59.2  | 65.5  | 66.9  | 65.3  | 58.6  | 50.3  | 41.6  | 34.2  |  |  |
|                                                                        | 10%         | DB     | 31.6  | 31.0  | 44.5  | 68.5  | 80.6  | 91.2  | 94.4  | 89.5  | 78.6  | 60.3  | 43.0  | 33.7  |  |  |
|                                                                        |             | MCWB   | 30.4  | 29.5  | 39.5  | 51.8  | 57.6  | 63.7  | 66.3  | 64.4  | 57.9  | 48.2  | 39.8  | 32.2  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 35.0  | 37.4  | 49.5  | 60.7  | 66.7  | 72.0  | 74.3  | 72.7  | 65.3  | 57.6  | 48.5  | 40.3  |  |  |
|                                                                        |             | MCDB   | 36.3  | 38.5  | 57.7  | 74.4  | 79.6  | 89.1  | 92.2  | 88.4  | 80.9  | 72.2  | 51.7  | 41.5  |  |  |
|                                                                        | 2%          | WB     | 33.0  | 33.3  | 45.4  | 57.7  | 63.6  | 69.5  | 71.8  | 70.0  | 62.6  | 54.6  | 45.2  | 37.0  |  |  |
|                                                                        |             | MCDB   | 33.8  | 34.1  | 51.7  | 70.6  | 81.0  | 85.6  | 90.0  | 86.7  | 80.0  | 65.6  | 48.1  | 38.3  |  |  |
|                                                                        | 5%          | WB     | 31.7  | 31.5  | 42.4  | 55.7  | 61.5  | 67.9  | 70.1  | 68.4  | 60.3  | 52.0  | 42.5  | 34.6  |  |  |
|                                                                        |             | MCDB   | 32.7  | 32.5  | 48.7  | 69.1  | 78.9  | 85.4  | 89.2  | 85.1  | 78.5  | 62.8  | 44.9  | 35.4  |  |  |
|                                                                        | 10%         | WB     | 30.2  | 29.6  | 39.6  | 53.3  | 59.6  | 66.5  | 68.7  | 66.9  | 58.3  | 49.4  | 39.6  | 32.5  |  |  |
|                                                                        |             | MCDB   | 31.6  | 30.9  | 44.5  | 66.5  | 76.2  | 83.9  | 87.0  | 83.7  | 75.5  | 58.9  | 42.6  | 33.4  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 12.8  | 14.8  | 14.8  | 20.9  | 21.9  | 21.7  | 21.4  | 21.2  | 22.1  | 20.1  | 12.5  | 11.1  |  |  |
|                                                                        |             | MCDBR  | 12.2  | 11.2  | 20.3  | 26.1  | 27.8  | 26.7  | 26.5  | 25.9  | 28.8  | 28.1  | 15.4  | 10.0  |  |  |
|                                                                        |             | MCWBR  | 10.8  | 10.1  | 12.7  | 12.0  | 12.1  | 10.1  | 10.7  | 10.4  | 13.6  | 15.6  | 11.1  | 8.9   |  |  |
|                                                                        | 5% WB       | MCDBR  | 11.2  | 10.0  | 18.6  | 22.7  | 23.7  | 22.1  | 21.4  | 21.6  | 25.9  | 24.4  | 12.5  | 8.9   |  |  |
|                                                                        |             | MCWBR  | 10.4  | 9.4   | 12.2  | 12.0  | 12.7  | 10.6  | 9.8   | 10.3  | 14.0  | 15.1  | 11.2  | 8.1   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.347 | 0.371 | 0.434 | 0.485 | 0.474 | 0.458 | 0.451 | 0.449 | 0.402 | 0.396 | 0.370 | 0.358 |  |  |
|                                                                        | taud        |        | 2.051 | 2.009 | 1.980 | 1.909 | 1.998 | 2.089 | 2.133 | 2.110 | 2.192 | 2.133 | 2.142 | 2.064 |  |  |
|                                                                        | Ebn at Noon |        | 218   | 241   | 244   | 244   | 253   | 258   | 258   | 252   | 253   | 230   | 209   | 196   |  |  |
|                                                                        | Edh at Noon |        | 31    | 40    | 48    | 57    | 54    | 49    | 47    | 46    | 39    | 35    | 29    | 28    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 359   | 666   | 1089  | 1571  | 1984  | 2213  | 2122  | 1883  | 1435  | 882   | 471   | 285   |  |  |
|                                                                        | RadStd      |        | 70    | 113   | 114   | 131   | 138   | 152   | 105   | 81    | 89    | 77    | 55    | 44    |  |  |

**Historical Trends**

|                             | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                             |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only           | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (2 neighbors) | N/A   | N/A     | N/A | N/A    | +2.39   | N/A   | -1.31 | N/A         | N/A   | +416  | +279  |

Nomenclature: See separate page